



Introduction

Entanglement is a fundamental component of quantum many-body systems at zero temperature. The characterization of such quantity essentially determines the phase order in which the system can be prepared. Therefore, its examination becomes crucial if we desire to distinguish the different phases of matter. Of special interest for this work are the topologically ordered phase, which are characterized by the presence of a topological contribution to the entanglement entropy [1, 2, 3]

$$S(\rho) \equiv -\text{Tr}(\rho \log \rho) \approx -\gamma + \alpha L + \mathcal{O}\left(\frac{1}{L}\right), \quad (1)$$

Namely, given a partition of the system, we expect the interior and exterior degrees of freedom to have a non-trivial entanglement. As a result, the intrinsic topological nature of the state ρ must be captured by a boundary independent term in $S(\rho)$, which is robust under deformations of such partition. This γ contribution is referred to as *Topological Entanglement Entropy (TEE)*. Furthermore, it can be shown that

$$\gamma = \log \mathcal{D}, \quad (2)$$

where $\mathcal{D} \equiv \sum_a d_a^2$ is the *total quantum dimension* of the model.

Sequential Quantum Circuits as Maps between Gapped Phases

Sequential Circuit [4, 5], more specially linear-depth circuits, are defined by the specific order in which a finite patch of unitary gates are applied. It is interesting to note that finite-depth circuits are a subclass of sequential circuits. This locally ordered feature of linear-depth circuits allows one to modify the entanglement of certain state, while preserving the gap. Consequently, this raises the question if the TEE of a state could be determined by simply looking at the sequential circuit that trivializes it.

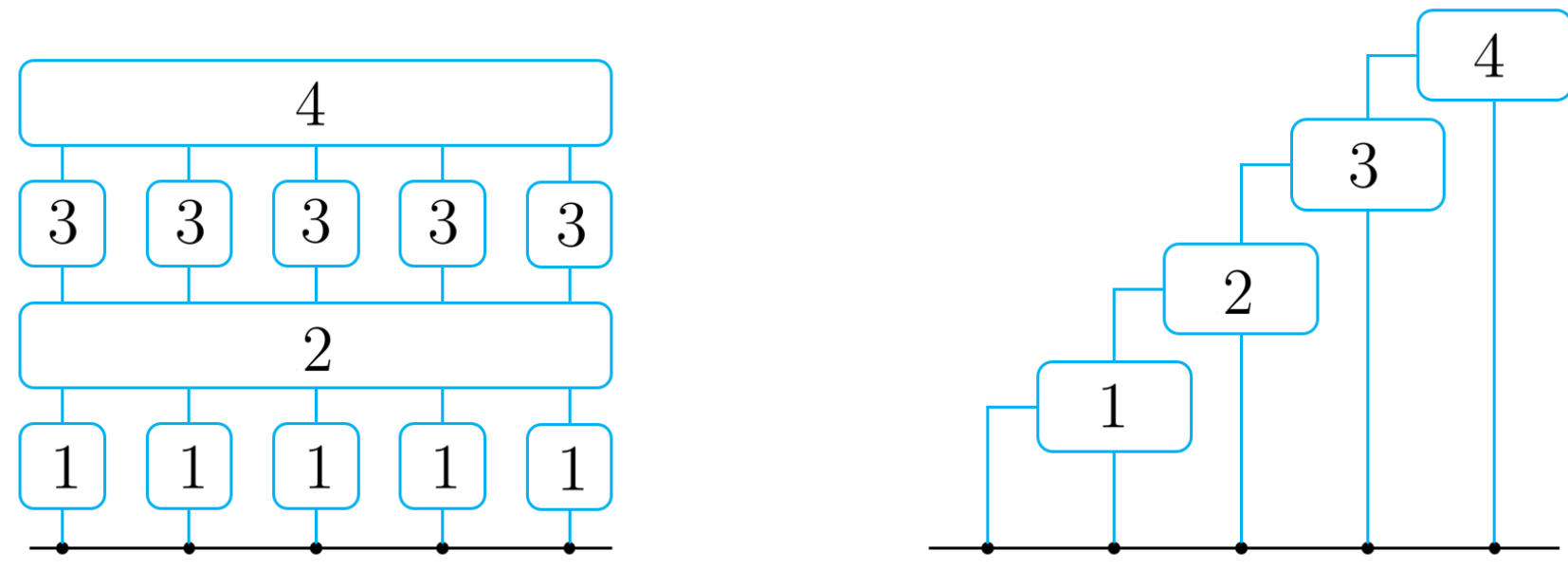


Figure 1. Example of a finite-depth circuit (left) and a linear-depth circuit (right).

Anyon condensation is a non-invertible process

The Toric code [6] is one of the canonical examples of a topological phase of matter, its ground state degeneracy is protected by topology and can be utilized to store quantum information in a way robust under local perturbation. The model is define in a square lattice as

$$H = -\sum_x \text{---} \text{---} \text{---} - \sum_z \text{---} \text{---} \text{---} \text{---}$$

$$A_s = x \text{---} \text{---} \text{---} x \quad B_p = z \text{---} \text{---} \text{---} z$$

This model also enjoys from an anomalous $\mathbb{Z}_2^{(e)} \times \mathbb{Z}_2^{(m)}$ symmetry group generated by the line operators

$$W(\gamma) = \prod_{j \in \gamma} Z_j, \quad T(\hat{\gamma}) = \prod_{j \in \hat{\gamma}} X_j,$$

$$W(\gamma)T(\hat{\gamma}) = (-1)^{\#\gamma \cap \hat{\gamma}} T(\hat{\gamma})W(\gamma).$$

Having set the stage with this brief introduction, we now consider the following composition of linear-depth circuits and global projections along a closed loop [5, 7]

Non-invertible boundary defects

$$\mathcal{D}_e(\gamma) = \sqrt{2} \left(\frac{1 + W(\gamma)}{2} \right) \left[\prod_{j=1}^{L-1} \left(\frac{1 + iZ_j}{\sqrt{2}} \right) \left(\frac{1 + iA_{s_j}}{\sqrt{2}} \right) \right] \left(\frac{1 + iZ_L}{\sqrt{2}} \right), \quad (3)$$

$$\mathcal{D}_m(\hat{\gamma}) = \sqrt{2} \left(\frac{1 + T(\hat{\gamma})}{2} \right) \left[\prod_{j=1}^{L-1} \left(\frac{1 + iX_j}{\sqrt{2}} \right) \left(\frac{1 + iB_{p_j}}{\sqrt{2}} \right) \right] \left(\frac{1 + iX_L}{\sqrt{2}} \right), \quad (4)$$

$$\mathcal{W}(\gamma) = W(\gamma) \prod_{j=1}^L A_{s_j}, \quad \mathcal{D}_e(\gamma) A_{s_j} = Z_{j+1} \mathcal{D}_e(\gamma), \quad \mathcal{D}_e(\gamma) \times \mathcal{D}_e^\dagger(\gamma) = \left(1 + W(\gamma) \right), \quad (5)$$

$$\mathcal{T}(\hat{\gamma}) = T(\hat{\gamma}) \prod_{j=1}^L B_{p_j}, \quad \mathcal{D}_m(\hat{\gamma}) B_{p_j} = X_{j+1} \mathcal{D}_m(\hat{\gamma}), \quad \mathcal{D}_m(\hat{\gamma}) \times \mathcal{D}_m^\dagger(\hat{\gamma}) = \left(1 + T(\hat{\gamma}) \right). \quad (6)$$

These electric and magnetic non-invertible operators are engineered such that the degrees of freedom along their curves are decoupled, a fact that we can verify by examining their action on the Toric code hamiltonian

$$\mathcal{D}_e(\gamma) H = \left(-\sum_{j \notin \gamma} A_{s_j} - \sum_p B_p - \sum_{j \in \gamma} Z_j \right) \mathcal{D}_e(\gamma), \quad (7)$$

$$\mathcal{D}_m(\hat{\gamma}) H = \left(-\sum_s A_s - \sum_{j \notin \hat{\gamma}} B_{p_j} - \sum_{j \in \hat{\gamma}} X_j \right) \mathcal{D}_m(\hat{\gamma}). \quad (8)$$

Or in other words, the topological state is transformed into a trivial product state along these defects.

Contractible Loops

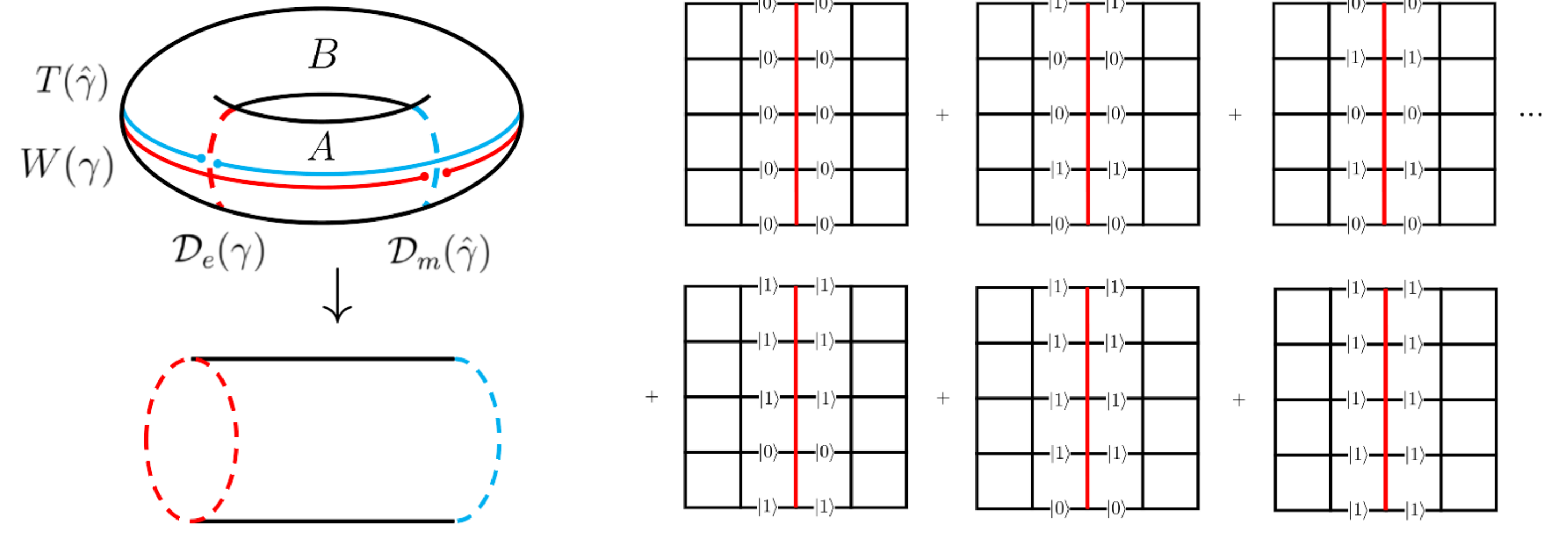


Figure 2. Bipartition of a torus using mixed boundary conditions implemented by $\mathcal{D}_e(\gamma)$ and $\mathcal{D}_m(\hat{\gamma})$, as well as the allowed boundary state after projection into the even sector of $\mathcal{W}(\gamma)$.

The existence of these non-invertible boundary defects allow us to consistently take a partition of the system, given that their action essentially creates a condensing boundary. One can therefore construct the reduced density matrix by employing their use as follows

$$\rho_A = \frac{1}{4} \text{Tr}_B \left(\mathcal{D}_e^\dagger(\gamma) \mathcal{D}_m^\dagger(\hat{\gamma}) \rho_{TC} \mathcal{D}_m(\hat{\gamma}) \mathcal{D}_e(\gamma) \right), \quad \rho_{TC} = \prod_s \left(\frac{1 + A_s}{2} \right) \prod_p \left(\frac{1 + B_p}{2} \right), \quad (9)$$

$$= \text{Tr}_B \left(\left[\frac{1 + W(\gamma)}{2} \right] \left[\frac{1 + T(\hat{\gamma})}{2} \right] \left(\rho_A \prod_{j \in \gamma} \left(\frac{1 + Z_j}{2} \right) \prod_{j \in \hat{\gamma}} \left(\frac{1 + X_j}{2} \right) \rho_B \right) \right), \quad (10)$$

where we have used the fusion rules of $\mathcal{D}_e(\gamma)$ and $\mathcal{D}_m(\hat{\gamma})$. Projecting in the even sectors of the dressed line operators $\mathcal{W}(\gamma)$ and $\mathcal{T}(\hat{\gamma})$ has the net effect of entangling the spins adjacent to the boundary. Furthermore, given that the resulting state is trivial along the boundary, the non-trivial contributions to the entanglement entropy can be phrased in terms of these global projections

$$S(\rho_B) = S \left(\text{Tr}_B \left[\frac{1 + W(\gamma)}{2} \right] \right) + S \left(\text{Tr}_B \left[\frac{1 + T(\hat{\gamma})}{2} \right] \right) \quad (11)$$

$$= 2^{L-1} \left(\frac{1}{2^{L-1}} \log \frac{1}{2^{L-1}} + \frac{1}{2^{L-1}} \log \frac{1}{2^{L-1}} \right) \Rightarrow \gamma = 2 \log 2, \quad (12)$$

where we have used the fact that the choice of mixed boundaries implies in an unique ground state on the remaining subsystem, once both $W(\gamma) = T(\hat{\gamma}) = 1$, and that each projection produces a coherent sum of 2^{L-1} spin state, given that only an even number of bonds crossing the boundary can be excited.

Non-contractible

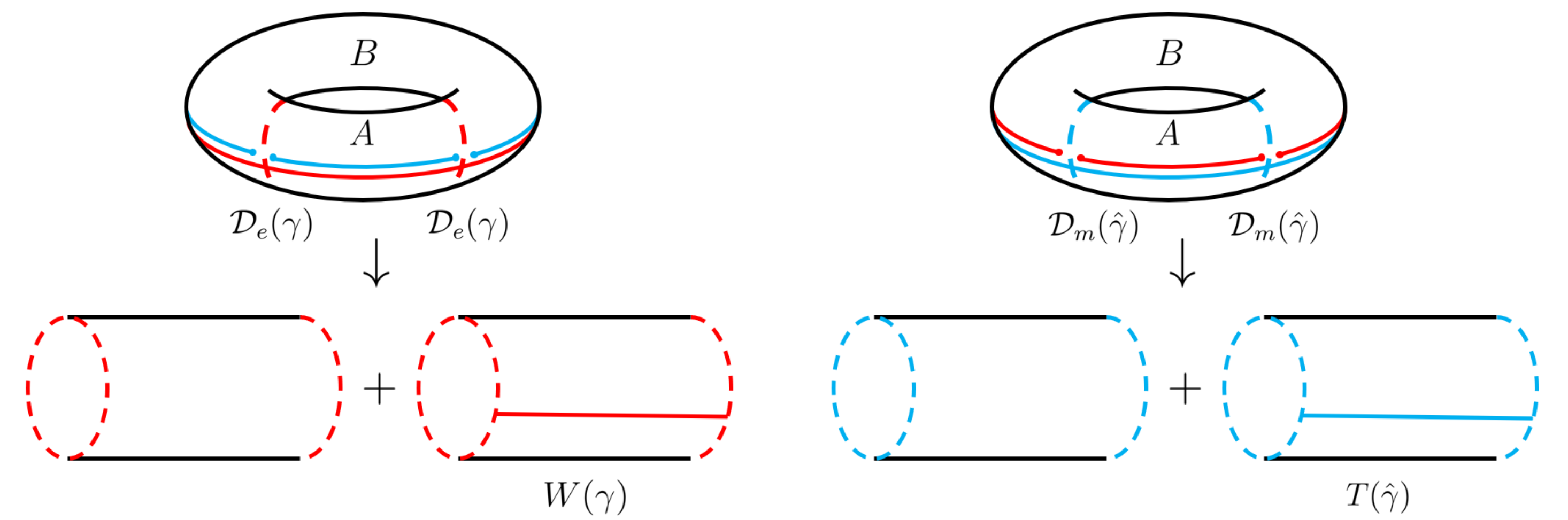


Figure 3. Matching partitions of the torus and the respective state in a cylindrical geometry.

Similarly, if we choose matching boundary conditions, the final cylindrical geometry possess a two-fold degeneracy, labeled by the quantum number $T(\hat{\gamma}) = \pm 1$ or $W(\gamma) = \pm 1$, respectively. Hence, when studying the mixed state

$$|TC\rangle = \sum_{w_x, w_y = \pm 1} c_{w_x w_y} |w_x w_y\rangle, \quad \rho_{TC} = |TC\rangle \langle TC|, \quad (13)$$

where $w_{x,y}$ denote the eigenvalues of the non-contractible Wilson loops, one needs to differentiate the even and odd sectors of the line operators while performing the chosen projections. In this set up both $\mathcal{W}(\gamma)$ and $\mathcal{T}(\hat{\gamma})$ pick an extra plus or minus sign depending in which sector we are at. As a result, the entanglement entropy becomes dependent on how the probabilities are distributed across sectors

$$S(\rho_A) = S \left(\text{Tr}_B \left[\frac{1 + W(\gamma)}{2} \right] \right) = (L-1) \log 2 - p_+ \log p_+ - p_- \log p_-. \quad (14)$$

Thus reproducing the result found at [3].

Conclusion

It is believed that this approach on the TEE can be easily generalized to a broad and interesting class of problems, e.g. thermal states, non-abelian anyons and fractons, where the determination of the entanglement entropy can prove itself a more convoluted task. Moreover, this approach could provide more insights in the study of area-to-volume law transitions.

References

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